A SHARP LOWER-BOUND CONSTRUCTION FOR ORDERED HYPERGRAPH MATCHINGS ALONG A FIXED CHAIN OF ORDERED PARTITIONS

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Allowed standard prerequisites (editorial).

- The construction of an r-partite ordered hypergraph matching with prescribed clique-size bounds. The author's lemma at source lines 556-560, whose proof at source lines 564-571 is delivered by this lane's companion card for that statement. Supplies the building block that the theorem iterates along the chain of ordered partitions.
- The quantity L_P of a collection and its role in the clique-size bounds. Defined in the author's preliminaries (source lines 225-281) and used through the display labelled Lp at source line 549. Names the quantities that the construction keeps below the prescribed bounds.

Source reference. Original equation 4.1: sources/combinatorics-fl02-a-sharp-ramsey-theorem-for-ordered/pinned-original-source.tex, source lines 549–549. Navigation

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pointer only: it labels the author's display defining the quantity L_P; the display belongs to the surrounding section and is not reproduced in this card.

Source reference. Original lemma 4.2: sources/combinatorics-fl02-a-sharp-ramsey-theorem-for-ordered/pinned-original-source.tex, source lines 556–556. Navigation pointer only: the counted proof iterates this construction lemma; it has its own card in this lane and its text is declared here as a named standard prerequisite.

Theorem 0.1. *Let r be a positive integer, and let $\lambda^{(1)} \succ \lambda^{(2)} \succ \cdots \succ \lambda^{(r)}$ be a sequence of ordered partitions of r. For every r-pattern $P \in \mathcal{P}(\lambda^{(1)}) \cup \cdots \cup \mathcal{P}(\lambda^{(r)})$, consider a positive integer m_P. Then there exists an ordered r-uniform hypergraph matching $\mathcal{H}$ of size*

$$|\mathcal{H}| = \prod_{P \in \mathcal{P}(\lambda^{(1)}) \cup \cdots \cup \mathcal{P}(\lambda^{(r)})} m_P,$$

such that any two edges $e, f \in \mathcal{H}$ form an r-pattern in $\mathcal{P}(\lambda^{(1)}) \cup \cdots \cup \mathcal{P}(\lambda^{(r)})$, and for each r-pattern $P \in \mathcal{P}(\lambda^{(1)}) \cup \cdots \cup \mathcal{P}(\lambda^{(r)})$ every P-clique contained in $\mathcal{H}$ has size at most m_P.

Proof of Theorem 0.1. We prove the theorem by induction on r. Note that the base case $r = 1$ is trivial, so let us assume $r \geq 2$ and that we already proved the theorem for $r - 1$.

In the sequence $\lambda^{(1)} \succ \lambda^{(2)} \succ \cdots \succ \lambda^{(r)}$, each ordered partition $\lambda^{(s)}$ for $s = 1, \ldots, r$ must have precisely s parts. In particular, the partition $\lambda^{(r-1)}$ has the form $\lambda^{(r-1)} = (1, \ldots, 1, 2, 1, \ldots, 1)$. Let $j \in \{1, \ldots, r-1\}$ be the unique index such that $\lambda_j^{(r-1)} = 2$. If we imagine $\lambda^{(1)} \succ \lambda^{(2)} \succ \cdots \succ \lambda^{(r)}$ as partitions of the set $\{1, \ldots, r\}$ into intervals, then the partition $\lambda^{(r-1)}$ consists of the set $\{j, j+1\}$ and singleton sets for all the remaining elements. Hence j and $j+1$ are in the same set for each of the partitions $\lambda^{(1)} \succ \lambda^{(2)} \succ \cdots \succ \lambda^{(r-1)}$. If we ignore j and relabel the remaining elements accordingly (i.e. $j+1$ is relabeled j, $j+2$ is relabeled $j+1$, and so on), $\lambda^{(1)} \succ \lambda^{(2)} \succ \cdots \succ \lambda^{(r-1)}$ turns into a sequence of ordered partitions $\mu^{(1)} \succ \mu^{(2)} \succ \cdots \succ \mu^{(r-1)}$ of $r-1$. For $s = 1, \ldots, r-1$ we obtain $\lambda^{(j)}$ from $\mu^{(j)}$ by doubling the element j (keeping both copies in the same set of the partition), and relabeling the ground set into $\{1, \ldots, r\}$ accordingly.

Now, for $s = 1, \ldots, r-1$, every r-pattern $P \in \mathcal{P}(\lambda^{(s)})$ turns into an $(r-1)$-pattern $P' \in \mathcal{P}(\mu^{(s)})$ when omitting the j-th A and the j-th B in P (note that these occur in the same block, i.e. right adjacent to the $(j+1)$-th A and the $(j+1)$-th B, respectively). Conversely, P can be recovered from $P' \in \mathcal{P}(\mu^{(s)})$ by doubling the j-th A in P into AA and doubling the j-th B in P into BB. Thus, these operations give natural bijections between $\mathcal{P}(\lambda^{(s)})$ and $\mathcal{P}(\mu^{(s)})$.

For every $(r-1)$-pattern $P' \in \mathcal{P}(\mu^{(1)}) \cup \cdots \cup \mathcal{P}(\mu^{(r-1)})$, let us define $m_{P'} = m_P$, where $P \in \mathcal{P}(\lambda^{(1)}) \cup \cdots \cup \mathcal{P}(\lambda^{(r-1)})$ is the r-pattern obtained from P' by doubling the j-th A and the j-th B. Then, applying the induction hypothesis to the sequence $\mu^{(1)} \succ \mu^{(2)} \succ \cdots \succ \mu^{(r-1)}$ of ordered partitions of $r-1$, we obtain an ordered $(r-1)$-uniform hypergraph matching $\mathcal{H}'$ of size

$$|\mathcal{H}'| = \prod_{P' \in \mathcal{P}(\mu^{(1)}) \cup \cdots \cup \mathcal{P}(\mu^{(r-1)})} m_{P'} = \prod_{P \in \mathcal{P}(\lambda^{(1)}) \cup \cdots \cup \mathcal{P}(\lambda^{(r-1)})} m_P,$$

such that $L_{P'}(\mathcal{H}') \leq m_{P'}$ for all $(r-1)$-patterns $P' \in \mathcal{P}(\mu^{(1)}) \cup \cdots \cup \mathcal{P}(\mu^{(r-1)})$ and $L_{P'}(\mathcal{H}') = 1$ for all other $(r-1)$-patterns $P' \notin \mathcal{P}(\mu^{(1)}) \cup \cdots \cup \mathcal{P}(\mu^{(r-1)})$.

From $\mathcal{H}'$, we can now obtain an ordered r-uniform hypergraph matching $\mathcal{H}^*$ with $|\mathcal{H}^*| = |\mathcal{H}'|$ by doubling the j-th element of every edge $e' \in \mathcal{H}'$. More formally, for every edge $e' \in \mathcal{H}'$, we add a new element to e' right before the j-the element of e' (in the given ordering of the underlying vertex set of $\mathcal{H}'$). For the resulting

edge $e \in \mathcal{H}^*$, we can recover the original edge $e' \in \mathcal{H}'$ by deleting the j-the element of e. If $e, f \in \mathcal{H}^*$ form some r-pattern P, then the $(r-1)$-pattern formed by the corresponding edges $e', f' \in \mathcal{H}'$ is the $(r-1)$-pattern P' obtained from P by omitting the j-th A and the j-th B. Thus, by the properties of $\mathcal{H}'$, we have $L_P(\mathcal{H}) \leq m_P$ for all r-patterns $P \in \mathcal{P}(\lambda^{(1)}) \cup \cdots \cup \mathcal{P}(\lambda^{(r-1)})$ and $L_P(\mathcal{H}) = 1$ for all other r-patterns $P \notin \mathcal{P}(\lambda^{(1)}) \cup \cdots \cup \mathcal{P}(\lambda^{(r-1)})$.

Finally, recall that $\lambda^{(r)} = (1, \ldots, 1)$ is the unique ordered partition of r into r parts. Let $\mathcal{H}_r$ be an r-partite ordered r-uniform hypergraph matching as in Lemma 4.2. Now, the blow-up $\mathcal{H} = \mathcal{H}^*[\mathcal{H}_r]$ has size

$$|\mathcal{H}| = |\mathcal{H}^*| \cdot |\mathcal{H}_r| = |\mathcal{H}'| \cdot |\mathcal{H}_r| = \prod_{P \in \mathcal{P}(\lambda^{(1)}) \cup \cdots \cup \mathcal{P}(\lambda^{(r-1)})} m_P \cdot \prod_{P \in \mathcal{P}(\lambda^{(r)})} m_P = \prod_{P \in \mathcal{P}(\lambda^{(1)}) \cup \cdots \cup \mathcal{P}(\lambda^{(r)})} m_P,$$

and by (4.1) for every r-pattern P we have $L_P(\mathcal{H}) \leq L_P(\mathcal{H}^*) \cdot L_P(\mathcal{H}_r)$. Hence $L_P(\mathcal{H}) \leq m_P \cdot 1 = m_P$ for all $P \in \mathcal{P}(\lambda^{(1)}) \cup \cdots \cup \mathcal{P}(\lambda^{(r-1)})$, and $L_P(\mathcal{H}) \leq 1 \cdot m_P = m_P$ for all $P \in \mathcal{P}(\lambda^{(r)})$, and also $L_P(\mathcal{H}) \leq 1 \cdot 1 = 1$ for all $P \notin \mathcal{P}(\lambda^{(1)}) \cup \cdots \cup \mathcal{P}(\lambda^{(r)})$. Thus, we have constructed the desired ordered r-uniform hypergraph matching $\mathcal{H}$. □

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